In the absence of external forces,

$$\mathbf{0} = \frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{l} = \dot{\boldsymbol{\Theta}}\boldsymbol{\pi} + \boldsymbol{\Theta}\dot{\boldsymbol{\pi}}.$$

Therefore

$$\dot{\boldsymbol{\pi}} = -\boldsymbol{\Theta}^T\dot{\boldsymbol{\Theta}}\boldsymbol{\pi}.$$

Next observe that

$$\boldsymbol{\Theta}^T\dot{\boldsymbol{\Theta}} = \mathrm{skew}(\boldsymbol{\omega}_\mathrm{b}).$$

This follows from the definition of the angular velocity and properties of the cross product (see exercises). Thus

$$\dot{\boldsymbol{\pi}} = -\boldsymbol{\omega}_\mathrm{b} \times \boldsymbol{\pi} = \boldsymbol{\pi} \times \boldsymbol{\omega}_\mathrm{b},$$

but since $\boldsymbol{\omega}_\mathrm{b} = \boldsymbol{T}^{-1}\boldsymbol{\pi}$, we may conclude

$$\dot{\boldsymbol{\pi}} = \boldsymbol{\pi} \times \boldsymbol{T}^{-1}\boldsymbol{\pi}. \tag{4.41}$$

These are the so-called *Euler equations* for the free rigid body.

Let us observe that (4.41) is in the form

$$\dot{\boldsymbol{\pi}} = \tilde{\boldsymbol{J}}(\boldsymbol{\pi})\nabla\tilde{H},$$

where $\tilde{\boldsymbol{J}}(\boldsymbol{\pi}) = \mathrm{skew}(\boldsymbol{\pi})$ is obviously skew symmetric, and $\tilde{H} = \boldsymbol{\pi}^T\boldsymbol{T}^{-1}\boldsymbol{\pi}/2$ is the energy function. Thus the Euler equations are in the form of a Hamiltonian system on a three-dimensional phase space. The symplectic structure matrix is dependent on the phase variables, and it is degenerate: $\tilde{\boldsymbol{J}}\boldsymbol{\pi} = \boldsymbol{\pi} \times \boldsymbol{\pi} = \mathbf{0}$, signifying that total angular momentum is conserved: $\|\boldsymbol{\pi}\| = \text{constant}$. This is a generalization of our previous concept of *Hamiltonian system*. Much of the previously developed theory of Hamiltonian systems can be adapted to the setting of the free rigid body with analogous definitions of splitting method and symplectic integrators obtained by additive decomposition of the Hamiltonian.

It is possible to solve the Euler equations using an exact solution and to use this to design efficient integrators [68, 249, 382], but in the most common setting of molecular dynamics, where additional approximations will be need, we are usually justified in treating these equations using a more elementary procedure [258]. If $\boldsymbol{T}$ is a diagonal matrix, $\boldsymbol{T} = \mathrm{diag}(I_1, I_2, I_3)$, then we split the Euler equations and approximate the solution by successive rotations around the three axes of inertia,

i.e. we compose the flows of

$$\dot{\boldsymbol{\pi}} = \boldsymbol{\pi} \times \begin{bmatrix} T_1^{-1}\pi_1 \\ 0 \\ 0 \end{bmatrix}, \qquad \dot{\boldsymbol{\pi}} = \boldsymbol{\pi} \times \begin{bmatrix} 0 \\ T_2^{-1}\pi \\ 0 \end{bmatrix}, \qquad \dot{\boldsymbol{\pi}} = \boldsymbol{\pi} \times \begin{bmatrix} 0 \\ 0 \\ T_3^{-1}\pi \end{bmatrix},$$

Each of these is, it turns out, easily integrated as a planar rotation (see Exercise 7). This yields a simple, efficient algorithm for the free rigid body which is symplectic in a generalized sense.

4.4.2 Rigid Body Molecular Dynamics Algorithm

To develop an integrator for the complete system (including external forces), we invoke the splitting technique,[2] solving successively for the translational motion from the center of mass equations, integrating the torques due to the rotational interactions, then solving the Euler equations, and recovering the rotation matrix from the linear system

$$\dot{\boldsymbol{\Theta}} = \boldsymbol{\Theta}\, \mathrm{skew}(\boldsymbol{T}^{-1}\boldsymbol{\pi}).$$

This is the approach described in [112] and is implemented in several software packages.

The reliability of these methods is supported by the fact that the combined scheme is a symplectic method in the generalized sense of constrained dynamics.

The specific steps of the algorithm proposed in [112] are given here. In what follows, we denote the variables associated to the kth rigid body by $\boldsymbol{q}^k$ (position of center of mass), $\boldsymbol{\Theta}^k$ (3×3 rotation matrix), $\boldsymbol{p}^k$ (linear momentum vector), and $\boldsymbol{\pi}^k$ (angular momentum vector). Individual components are defined by double subscripts: the 2nd component of $\boldsymbol{\pi}^k$ is π_2^k. Procedures given below for rigid body k are meant to be performed for all rigid bodies in the system.

The algorithm is initialized by computing the forces $\boldsymbol{F}^k$ and torques $\boldsymbol{\tau}^k$ acting on the kth rigid body, evaluated at the initial point. The relevant expressions are:

$$\boldsymbol{F}^k = -\frac{\partial U}{\partial \boldsymbol{q}^k}, \qquad \boldsymbol{\tau}^k = -\mathrm{rot}\left(\left[\boldsymbol{\Theta}^k\right]^T \frac{\partial U}{\partial \boldsymbol{\Theta}^k} \right).$$

[2] We note the seminal role of the work in astronomy of Touma and Wisdom [368] in elucidating the "divide and conquer" approach for deriving such schemes for rigid bodies.